Syed Ibrahim Mustafa Shah Bukhari

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Education

Virginia Tech **Expected May 2028**

Ph.D. in Computer Science · *Thesis: Privacy in Extended Reality (XR)*

Virginia Tech **August 2023 – May 2026**

M.S. in Computer Science · *GPA: 4.00*

Lahore University of Management Sciences (LUMS) **August 2018 – June 2022**

B.S. in Computer Science · Minor in Psychology · *Major GPA: 3.92*

Research Experience

Graduate Research Assistant **August 2023 – Present**

Virginia Tech Blacksburg, VA

Advisors: Dr. Brendan David-John and Dr. Bo Ji

EvaluatAR: Reproducible Cross-Device Evaluation of Bystander Privacy-Enhancing Technologies (PETs) for AR

- Pinpointed that conventional bystander PETs' evaluations rely on live participants, incur substantial setup overhead, and remain device-specific, undermining reproducibility across headsets and slowing iterative development.
- Established a controlled early-stage evaluation protocol that enables researchers and practitioners to reliably assess system feasibility, robustness, and privacy-performance trade-offs under repeatable conditions.
- Engineered EvaluatAR, a modular record-replay framework developed using Unity and Python that standardizes sensor inputs, visual stimuli, runtime measurements, and system outputs across headsets.
- Demonstrated EvaluatAR's cross-device applicability across 2 PETs and 3 headsets, enabling one experimenter to execute 150 controlled trials in about 60 minutes and rapidly prototype and extensively validate fixes for privacy-relevant failures.
- **Impact:** First-author paper at *PoPETs'26*; artifact **Available** and **Functional** badges; CCI Research Paper Showcase'26.

Rethinking Privacy Indicators in Extended Reality (XR)

- Recognized that conventional XR privacy indicators assume bystanders are attentive and able to perceive visual cues, leaving distracted or otherwise situationally impaired bystanders unaware of nearby sensing and recording.
- Formulated an inclusive design objective to make XR privacy indicators perceivable to situationally impaired bystanders by adapting their modality, salience, and activation to the surrounding context.
- Led 4 participatory focus groups ($N = 8$), eliciting 7 concepts and consolidating them into 5 novel privacy-indicator designs spanning visual, auditory, haptic, proximity-, gaze-, and activity-triggered cues.
- Assessed the proposed designs through an 8-scenario user study ($N = 7$), finding all to exceed the accepted SUS threshold and multimodal indicators being preferred over visual-only cues in privacy-sensitive impairment scenarios.
- **Impact:** First-author paper at *ISMAR-Adjunct'25*.

"Just Stop Doing Everything for Now!": Security Attacks in Collaborative MR

- Exposed a gap in the security of remote collaborative MR, where users' real-time recognition and mitigation of immersive security attacks had not been studied in situ.
- Devised and refined a counterbalanced mixed-methods protocol through 2 pilot rounds, calibrating the collaborative task, attack conditions, behavioral measures, questionnaires, and interview guide.
- Orchestrated the IRB-approved study with 20 participants in 10 remote pairs, capturing task performance, workload, usability, security perceptions, behavioral recordings, and interviews across baseline and attack conditions.
- Triangulated quantitative, observational, and qualitative evidence, finding that click redirection increased average completion time from 5.25 to 8.54 minutes ($p = 0.02$), while 67% of participants exposed to spatial occlusion misreported their actual mitigation.
- **Impact:** Published at *IEEE VR'25*; featured by *Computerworld*.

Visceral Interfaces for Privacy Awareness of Eye Tracking in VR

- Investigated whether visceral interfaces could make otherwise invisible VR eye-tracking collection understandable to users without disrupting their experience.
- Designed a counterbalanced mixed-methods study protocol to evaluate 3 Meta Quest Pro interfaces using pre-/post-study privacy measures, scenario surveys, behavioral observations, simulator-sickness screening, and semi-structured interviews.
- Conducted IRB-approved study with 40 participants, enabling comparison of privacy attitudes, usability, and interface preferences.
- Synthesized quantitative and qualitative findings into design recommendations, finding that visceral interfaces significantly increased caution toward sharing gaze data with employers and VR entertainment applications ($p < 0.001$).
- **Impact:** Published at *ISMAR'24*.

Undergraduate Research Assistant **January 2022 – May 2023**
LUMS Lahore, Pakistan
Advisors: Dr. Muhammad Fareed Zaffar and Dr. Murtaza Taj

TaxData: Human-Centered Remote-Sensing Support for Property Tax Surveying

- Identified slow, manual property surveying as a barrier to tax assessment; curated training, validation, and test data from 30,000 Google Earth images, including 3,000 manually labeled constructed and non-constructed examples.
- Developed a remote-sensing pipeline using deep-learning classification and connected-component and YOLO-based plot-segmentation methods, achieving 92% constructed-versus-non-constructed classification accuracy.
- Translated the model into a full-stack decision-support application that enabled tax officials to define regions of interest and inspect current predictions, reducing property-survey time by 70%.
- **Impact:** Featured in *International Growth Centre (IGC March 2022)* technical report.

Publications

- "EvaluatAR: A Cross-Device Evaluation Framework for Rapid Prototyping of Bystander PETs in AR." **S. Bukhari**, M. Corbett, B. Ji, and B. David-John. *Proceedings on Privacy Enhancing Technologies (PoPETs 2026(4)).*
- "Rethinking Privacy Indicators in Extended Reality: Multimodal Design for Situationally Impaired Bystanders." **S. Bukhari**, M. Sajid, B. Ji, and B. David-John. *2025 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct 2025).*
- "Just Stop Doing Everything for Now!": Understanding Security Attacks in Remote Collaborative Mixed Reality." M. Sajid, **S. Bukhari**, B. Ji, and B. David-John. *2025 IEEE Conference on Virtual Reality and 3D User Interfaces (IEEE VR 2025).*
- "Visceral Interfaces for Privacy Awareness of Eye Tracking in VR." G. Ramirez-Saffy, P. Chelladurai, A. Vargas, **S. Bukhari**, E. Selinger, S. Foster, B. Heller, and B. David-John. *2024 IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2024).*

Additional Experience

Graduate Teaching Assistant **January 2025 – Present**
Virginia Tech Blacksburg, VA

- Courses: System and Software Security, Professionalism in Computing, and Introduction to Programming.

Undergraduate Teaching Assistant **September 2020 – August 2022**
LUMS

- Courses: Intro to Programming, AI, Software Engineering, Computer Vision Fundamentals, and Computational Problem Solving.

Full-Stack Software Engineer **March 2023 – August 2023**
Senarios

- Architected a cross-platform real-time contact-sharing application, authored design documents, and integrated decentralized components; launched on Google Play Store with active users.
- Advised 3+ clients on Web3-based data immutability while coordinating a cross-functional Agile team.

Backend Software Engineer **August 2022 – March 2023**
GoSaaS Labs

- Developed seven cloud-to-network-drive features, improving data retrieval speed by 15%.
- Optimized SQL queries to reduce backend response time by 25%.

Technical Skills

HCI and Research Methods: Mixed methods, controlled experiments, participatory design, focus groups, scenario-based studies, usability testing, within-subjects designs, qualitative and quantitative data analysis, statistical analysis
XR and Prototyping: Unity, OpenXR, MRTK, HoloLens 2, Magic Leap 2, Meta Quest 3, Figma
AI and Data: Python, R, PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib
Software Engineering: C++, C#, SQL, JavaScript, Node.js, Express.js, Git, Docker, Microsoft Azure, Amazon Web Services

Service & Awards

- **Mentoring:** Mentored Kim Giordano and Amulya Chinnala; UR2PhD-certified Graduate Student Mentor.
- **Reviewing:** *VRST'26*, and GenXR-ID workshop at *CVPR'26*.
- **Leadership & Service:** Virginia Tech Computer Science Graduate Council Webmaster (2024–25), Treasurer (2025–26), and Vice President (2026–27); iXR Ideathon organizing committee (2026).
- **Awards:** CCI Cyber Innovation Scholar (2 awards; $3,500 total), PoPETs'26 stipend ($840), and Dean's Honor List.